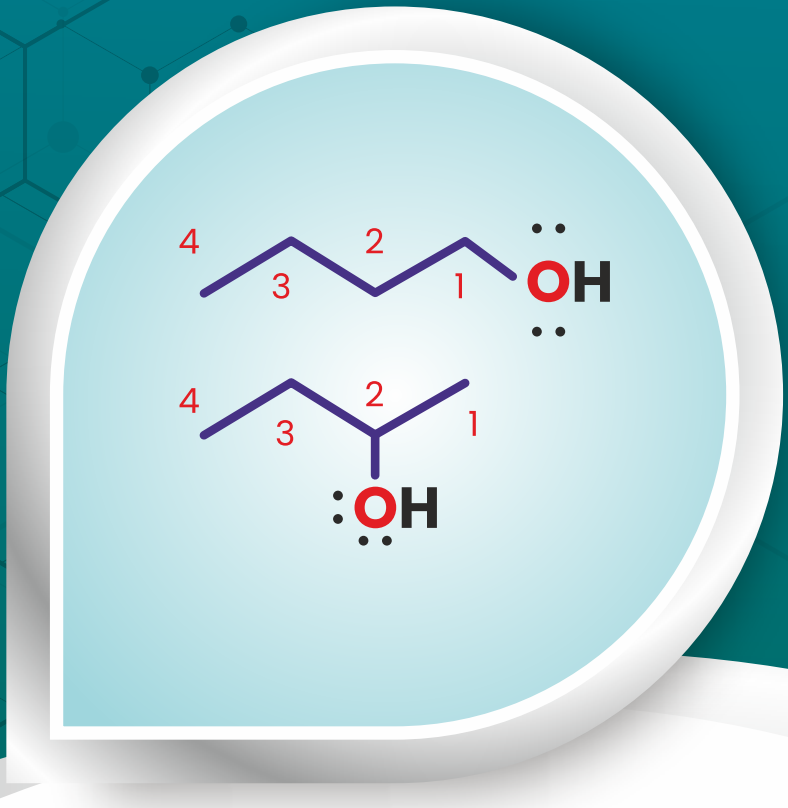
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

path to success

ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Nomenclature & Common Names

(English Medium)

NOMENCLATURE OF ORGANIC COMPOUND AND COMMON NAMES

EXERCISE # O-I

1. How many 1°carbon atom will be present in a simplest open chain hydrocarbon having two 3° and one 2° carbon atom ?

(A) 3 (B) 4 (C) 5 (D) 6

NC0001

2. Alicyclic compounds are :

(A) Aromatic compounds (B) Aliphatic cyclic compounds

(C) Heterocyclic compounds (D) None of the above

NC0002

3. How many 1°, 2°, 3° C atoms does 1, 3, 5-Trimethyl cyclohexane have?

(A) 3, 6, 0 (B) 3, 4, 2 (C) 0, 3, 6 (D) 3, 3, 3

NC0003

4. The compound which has one isopropyl group is:

(A) 2,2,3,3-Tetramethyl pentane (B) 2,2-Dimethyl pentane

(C) 2,2,3-Trimethyl pentane (D) 2-Methyl pentane

NC0004

5. Which of the following is the first member of ester homologous series?

(A) Ethyl ethanoate (B) Methyl ethanoate

(C) Methyl methanoate (D) Ethyl methanoate

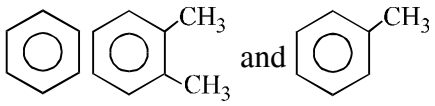
NC0005

6. A group closely related compounds which can be expressed by a general formula & in which two consecutive members differ by 14 in their molecular masses is called

(A) a heterogeneous series (B) a homologous series

(C) a homogeneous series (D) a electrochemical series

NC0006

7. 

Number of secondary carbon atoms present in the above compounds are respectively:

(A) 6,4,5 (B) 4,5,6 (C) 5,4,6 (D) 6,2,1

NC0007

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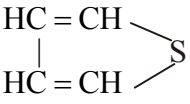
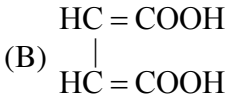
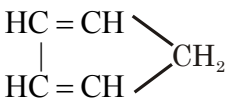
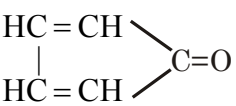
8. The molecular formula of the first member of the family of alkenynes and its name is given by the set

(A) C_3H_6, Alkene (B) C_5H_6, Pent-1-en-3-yne

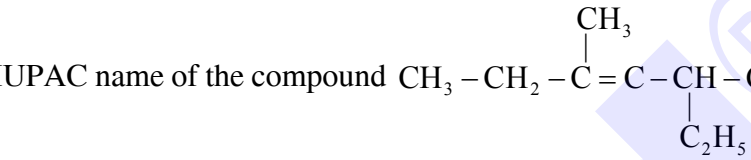
(C) C_6H_8, Hex-1-en-5-yne (D) C_4H_4, Butenyne

NC0008

9. Which of the following is a heterocyclic compound :

(A)  (B)  (C)  (D) 

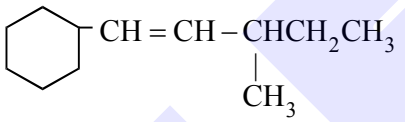
NC0009

10. The correct IUPAC name of the compound  :

(A) 5-Ethyl-3, 6-dimethyl non-3-ene (B) 5-Ethyl-4, 7-dimethyl non-3-ene

(C) 4-Methyl-5, 7-diethyl oct-2-ene (D) 2,4-Ethyl-5-methyl oct-2-ene

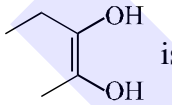
NC0010

11. The IUPAC name of  is:

(A) 1-Cyclohexyl-3-methyl pent-1-ene (B) 3-Methyl-5-cyclohexyl pent-1-ene

(C) 1-Cyclohexyl-3-ethyl but-1-ene (D) 1-Cyclohexyl-3,4-dmethyl but-1-ene

NC0011

12. IUPAC name of  is:

(A) But-2-ene-2,3-diol (B)Pent-2-ene-2,3-diol

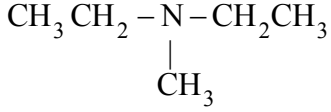
(C) 2-Methylbut-2-ene-2,3-diol (D) Pent-3-ene-3,4-diol

NC0012

13. IUPAC name of CH_2=CH – CN is:

(A) Ethenenitrile (B) Vinyl cyanide (C) Cyono ethene (D) Prop-2-enenitrile

NC0013

14. The IUPAC name of  is:

(A) N-Methyl-N-ethyl ethanamine (B) Diethyl methanamine

(C) N-Ethyl-N-methyl ethanamine (D) Methyl diethyl ethanamine

NC0014

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15. The IUPAC name of acetyl acetone is :

(A) Pentane-2,5- dione (B)Pentane -2,4-dione

(C) Hexane-2,4-dione (D)Butane-2,4-dione

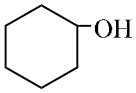
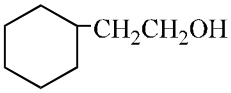
NC0015

16. When vinyl & allyl are joined each other, we get

(A) Conjugated alkadiene (B) cumulative alkadiene

(C) Isolated alkadiene (D) Allenes

NC0016

17. (a)  and (b) 

True statement for the above compounds is :

(A) (a) is phenol while (b) is alcohol (B) Both (a) and (b) are primary alcohol

(C) (a) is primary and (b) is secondary alcohol (D) (a) is secondary and (b) is primary alcohol

NC0017

18. The IUPAC name of the following structure $(CH_3)C.C.C.(CH_3)CH(CH_3)$ is:

(A) 3-Methylhex-4-yn-2-ene (B) 3-Methylhex-2-en-4-yne

(C) 4-Methylhex-4-en-4-yne (D) All are correct

NC0018

19. The IUPAC name of the following structure is $[CH_3CH(CH_3)]_2\ C(CH_2CH_3)C(CH_3)\ C(CH_2CH_3)_2$

(A) 3,5-Diethyl-4,6-dimethyl-5-[1-methylethyl]hept-3-ene

(B) 3,5-Diethyl-5-isopropyl-4,6-dimethylhept-2-ene

(C) 3,5-Diethyl-5-propyl-4,6-dimethylhept-3-ene

(D) None of these

NC0019

20. The correct IUPAC name of $CH_3-CH_2-\underset{\underset{CH_2}{||}}{C}-COOH$ is:

(A) 2-Methyl butanoic acid (B) 2-Ethylprop-2-enoic acid

(C) 2-Carboxybutene (D) None of the above

NC0020

21. The correct IUPAC name of 2-ethylpent-3-yne is:

(A) 3-Methyl hex-4-yne (B) 4-Ethyl pent-2-yne

(C) 4-methyl hex-2 yne (D) None of these

NC0021

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22. All the following IUPAC names are correct except:

(A) 1-Chloro-1-ethoxy propane (B) 1-Amino-1-ethoxypropane

(C) 1-Ethoxy-2-propanol (D) 1-Ethoxy-1-propanamine

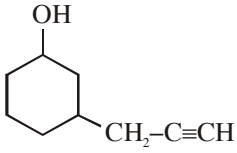
NC0022

23. The IUPAC name of the compound $CH_3CH = CHCH=CHC\equiv CCH_3$ is:

(A)Octa-4,6-diene-2-yne (B) Octa-2,4-diene-6-yne

(C) Oct-2-yne-4,6-diene (D) Oct-6-yne-2,4-diene

NC0023

24. The correct IUPAC name of 

(A) 3-Cyclohexanol Propyne (B) 3-[3-Hydroxy Cyclohexyl] Propyne

(C) 3-Propynyl Cyclohexanol (D) 3-(2-propynyl) Cyclohexanol

NC0024

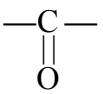
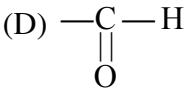
25. The IUPAC name of β-ethoxy-α-hydroxy propionic acid (trivial name) is:

(A) 1,2-Dihydroxy-1-oxo-3-ethoxy propane (B) 1-Carboxy-2-ethoxy ethanol

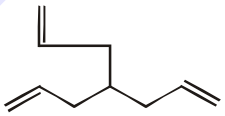
(C) 3-Ethoxy-2-hydroxy propanoic acid (D) All above

NC0025

26. As per IUPAC rules, which one of the following groups, will be regarded as the principal functional group ?

(A) —C≡C— (B) —OH (C)  (D) 

NC0026

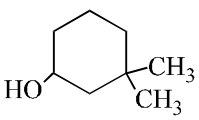
27. The IUPAC name of the compound  is :

(A) 4-Prop-1-enyl hepta-1,6-diene (B) 4-Propylidene hepta-1,6-diene

(C) 4-Propenyl hepta-1,6-diene (D) 4-[Prop-2-enyl] hepta-1,6-diene

NC0027

28. The IUPAC name of the given compound is:



(A) 1,1-Dimethyl-3-hydroxy cyclohexane (B) 3,3-Dimethyl-1-hydroxy cyclohexane

(C) 3,3-Dimethylcyclohexanol (D) 1,1-Dimethylcyclohexan-3-ol

NC0028

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29. The IUPAC name of $(C_2H_5)_2NCH_2\underset{\displaystyle Cl}{\underset{|}{C}}H.COOH$ is:

(A) 2-Chloro-4-N-ethylpentanoic acid

(B) 2-Chloro-3-(N,N-diethyl amino)-propanoic acid

(C) 2-Chloro-2-oxo diethylamine

(D) 2-Chloro-2-carboxy-N-ethyl ethane

NC0029

30. The IUPAC name of the compound is $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\underset{\displaystyle Ph}{\underset{|}{CH}}-NH_2$

(A) 1-Amino-1-phenyl-2-methyl propane

(B) 2-Methyl-1-phenyl propan-1-amine

(C) 2-Methyl-1-amino-1-phenyl propane

(D) 1-Isopropyl-1-phenyl methyl amine

NC0030

31. Which of the following compound is wrongly named ?

(A) $CH_3CH_2CH_2\underset{\displaystyle Cl}{\underset{|}{C}}HCOOH$; 2-Chloro pentanoic acid

(B) $CH_3C\equiv\underset{\displaystyle CH_3}{\underset{|}{C}}CHCOOH$; 2-Methyl hex-3-enoic acid

(C) $CH_3CH_2CH=CHCOCH_3$; Hex-3-en-2-one

(D) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}}HCH_2CH_2CHO$; 4-Methyl pentanal

NC0031

32. The correct IUPAC name of the following compound is:

$$O=\underset{\displaystyle OH}{\underset{|}{C}}-CH_2-\underset{\displaystyle H-C=O}{\underset{|}{CH}}-CHO$$

(A) 3,3-Diformylpropanoic acid

(B) 3-Formyl-4-oxo-butanoic acid

(C) 3,3-Dioxo propanoic acid

(D) 3,3-Dicarbaldehyde propanoic acid

NC0032

33. The correct IUPAC name of compound $CH_3-CH_2-\underset{\displaystyle O}{\underset{||}{C}}-\underset{\displaystyle CN}{\underset{|}{CH}}-CHO$ is :

(A) 2-Cyano-3-oxopentanal

(B) 2-Formyl-3-oxopentanenitrile

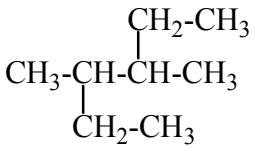
(C) 2-Cyanopentane-1,3-dione

(D) 1,3-Dioxo-2-cyanopentane

NC0033

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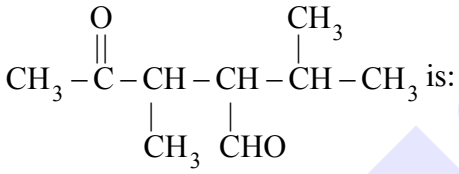
34. IUPAC name of compound



(A) 2, 3-diethyl butane (B) 2-ethyl-3-methyl pentane

(C) 3-methyl-2-ethyl pentane (D) 3,4-dimethyl hexane

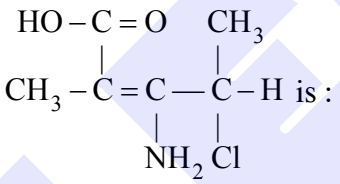
NC0034

35. The IUPAC name of compound  is:

(A) 3,5-Dimethyl-4-Formyl pentanone (B) 1-Isopropyl-2-methyl-4-oxo butanal

(C) 2-Isopropyl-3-methyl-4-oxo pentanal (D) None of the above

NC0035

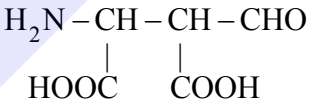
36. The IUPAC name of compound  is :

(A) 2-Amino-3-chloro-2-methylpent-2-enoic acid (B) 3-Amino-4-chloro-2-methylpent-2-enoic acid

(C) 4-Amino-3-chloro-2-methylpent-2-enoic acid (D) All of the above

NC0036

37. The IUPAC name of the structure is:

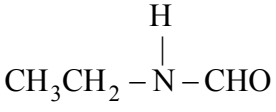


(A) 3-Amino-2-formyl butane-1, 4-dioic acid (B) 3-Amino-2, 3-dicarboxy propanal

(C) 2-Amino-3-formyl butane-1, 4-dioic acid (D) 1-Amino-2-formyl succinic acid

NC0037

38. One among the following is the correct IUPAC name of the compound



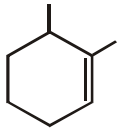
(A) N-Formyl aminoethane (B) N-Ethyl formyl amine

(C) N-Ethyl methanamide (D) Ethylamino methanal

NC0038

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39. The IUPAC name of the structure is :



(A) 1,2-Dimethyl-Cyclohexane
(B) 1,6-Dimethyl-Cyclohexene
(C) 1,2-Dimethyl-Cyclohex-2-ene
(D) 2,3-Dimethyl-Cyclohexane

NC0039

40. The IUPAC name of $C_6H_5CH=CH–COOH$ is :

(A) Cinnamic acid
(B) 1-Phenyl-2-carboxy ethane
(C) 3-Phenyl prop-2-enoic acid
(D) Dihydroxy-3-phenyl propionic acid

NC0040

41. The IUPAC name of $BrCH_2-\underset{\underset{CONH_2}{|}}{CH}-CO-CH_2-CH_2CH_3$ is:

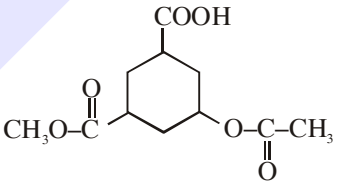
(A) 2-Bromomethyl-3-oxohexanamide
(B) 1-Bromo-2-amino-3-oxohexane
(C) 1-Bromo-2-amino-n-propyl ketone
(D) 3-Bromo-2-propyl propanamide

NC0041

42. IUPAC name will be $\underset{\underset{CN}{|}}{CH_2}-\underset{\underset{CN}{|}}{CH}-\underset{\underset{CN}{|}}{CH_2}$

(A) 1,2,3-Tricyano propane
(B) Propane-1,2,3- trinitrile
(C) 1,2,3-Cyano propane
(D) Propane-1,2,3-tricarbonitrile

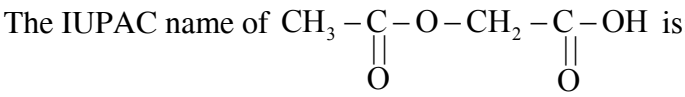
NC0042

43. The IUPAC name of compound 

(A) 3-Carbonyl methoxy -5- Ethanoyl oxy cyclohexanoic acid
(B) 3-Ethanoyl oxy -5- Methoxy carbonyl cyclohexane carboxylic acid
(C) 5-Ethanoyl oxy -5- Methoxy carbonyl cyclohexanoic acid
(D) 3-Methoxy carbonyl -5- Ethanoyl oxy cyclohexane carboxylic acid

NC0043

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44. The IUPAC name of  is:

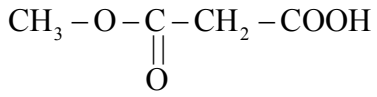
(A) 1-Acetoxy acetic acid

(B) 2-Acetoxy ethanoic acid

(C) 2-Ethanoyloxyacetic acid

(D) 2-Ethanoyloxyethanoic acid

NC0044

45. 

The correct IUPAC systematic name of the above compound is:

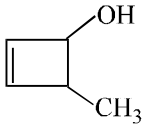
(A) 2-Acetoxy ethanoic acid

(B) 2-Methoxy carbonyl ethanoic acid

(C) 3-Methoxy formyl ethanoic acid

(D) 2-Methoxy formyl acetic acid

NC0045

46. The IUPAC name of  is :

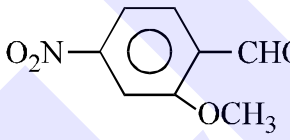
(A) 3-Methyl cyclobut-1-ene-2-ol

(B) 4-Methyl cyclobut-2-ene-1-ol

(C) 4-Methyl cyclobut-1-ene-3-ol

(D) 2-Methyl cyclobut-3-ene-1-ol

NC0046

47. The IUPAC name of  is:

(A) 2-Methoxy-4-nitro benzaldehyde

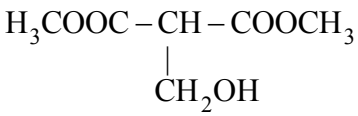
(B) 4-Nitro anisaldehyde

(C) 3-Methoxy-4-formyl nitro benzene

(D) 2-Formyl-4-nitro anisole

NC0047

48. The IUPAC name of compound



(A) 2-(Hydroxy methyl) methyl propanedioate

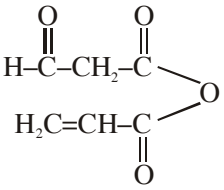
(B) Methyl-2-(hydroxy methyl) propanedioate

(C) 2-(Hydroxy methyl) dimethyl propanedioate

(D) None of these

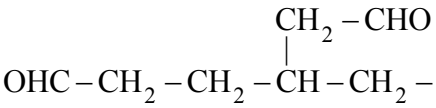
NC0048

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3.1Nomen-EDE-A & B\Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

49. The IUPAC name of 

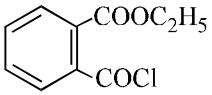
(A) 2-Formyl ethanoic propanoic Anhydride
(B) 2-Oxo-propanoic prop-2-enoic Anhydride
(C) Prop-2-enoic-2-formyl propanoic Anhydride
(D) 2-Formyl ethanoic prop-2-enoic Anhydride

NC0049

50. The IUPAC name of  is:

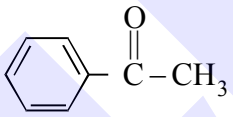
(A) 4,4-Di(formylmethyl) butanal
(B) 2-(Formylmethyl) butane-1, 4-dicarbaldehyde
(C)Hexane-3-acetal-1, 6-dial
(D) 3-(Formylmethyl) hexane-1, 6-dial

NC0050

51. The IUPAC name of  is :

(A) 2-Chlorocarbonyl ethylbenzoate
(B) 2-Carboxyethyl benzoyl chloride
(C) Ethyl-2-(chlorocarbonyl)benzoate
(D) Ethyl-1-(chlorocarbonyl)benzoate

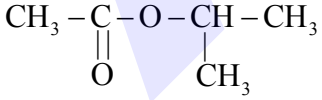
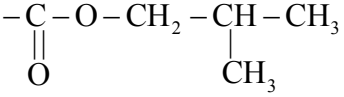
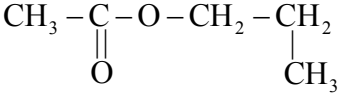
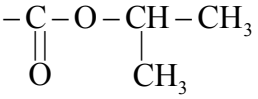
NC0051

52. The IUPAC name of  is:

(A) Phenyl ethanone
(B) Methyl phenyl ketone
(C) Acetophenone
(D) Phenyl methyl ketone

NC0052

53. Structural formula of isopropyl methanoate is :

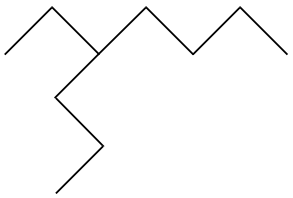
(A) 
(B) 
(C) 
(D) 

NC0053

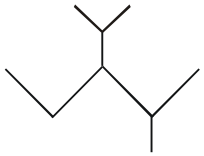
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3-Nomen-EDE-A & B\Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

EXERCISE # O-II

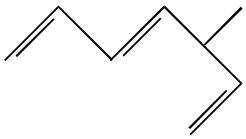
Give the IUPAC names for each of the following :

1. 

NC0054

2. 

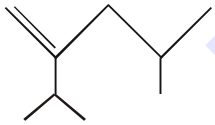
NC0055

3. 

NC0056

4. 

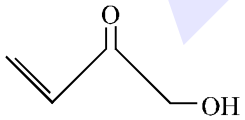
NC0057

5. 

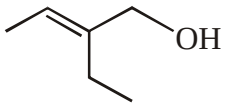
NC0058

6. 

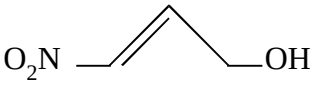
NC0059

7. 

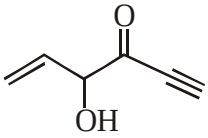
NC0060

8. 

NC0061

9. 

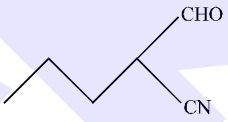
NC0062

10. 

NC0063

11. 

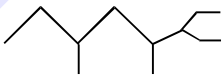
NC0064

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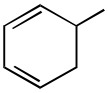
NC0065

13. 

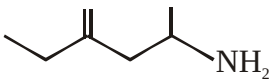
NC0066

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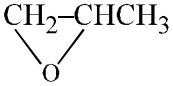
NC0067

15. 

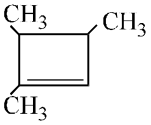
NC0068

16. 

NC0069

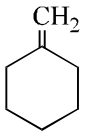
17. 

NC0070

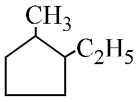
18. 

NC0071

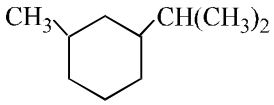
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3-Nomen-EDE-A & B\Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

19. 

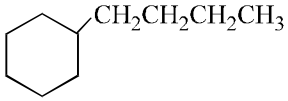
NC0072

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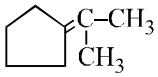
NC0073

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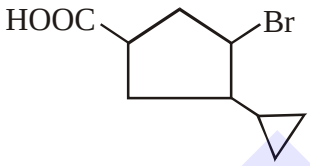
NC0074

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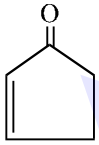
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23. 

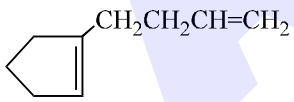
NC0076

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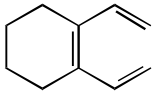
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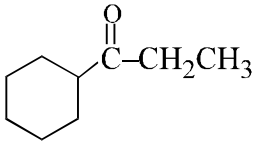
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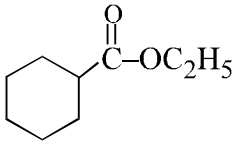
NC0079

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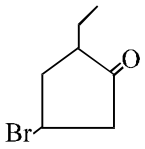
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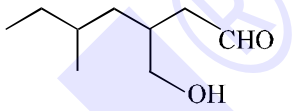
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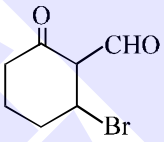
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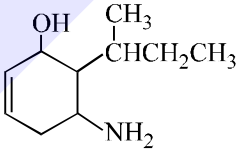
NC0083

31. 

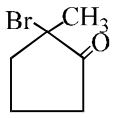
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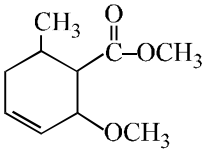
NC0085

33. 

NC0086

34. 

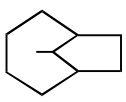
NC0087

35. 

NC0088

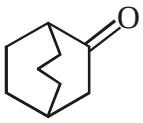
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NC0089

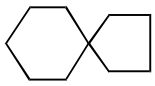
37. 

NC0090

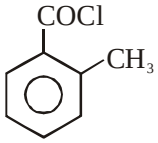
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3-Nomen-EDE-A & B\Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

38. 

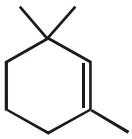
NC0091

39. 

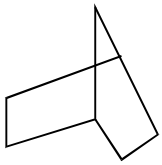
NC0092

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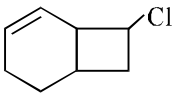
NC0093

41. 

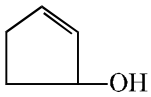
NC0094

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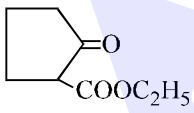
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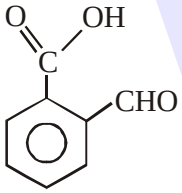
NC0096

44. 

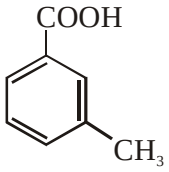
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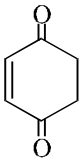
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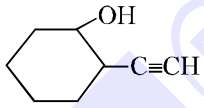
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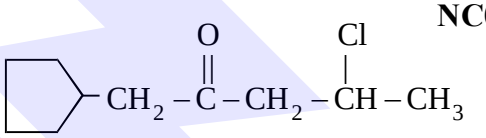
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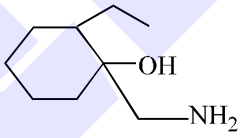
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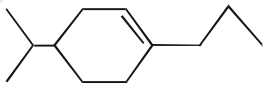
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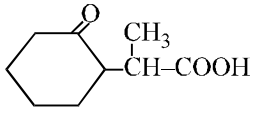
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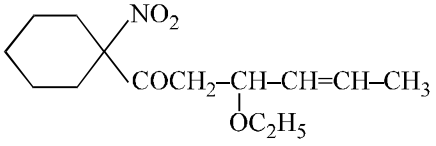
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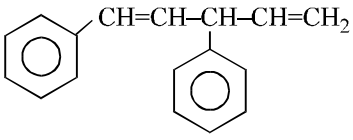
NC0105

53. 

NC0106

54. 

NC0107

55. 

NC0108

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EXERCISE # S-I

1. Which of the following pairs have absence of carbocyclic ring in both compounds?

(A) Pyridine, Benzene (B) Benzene, Cyclohexane

(C) Cyclohexane, Furane (D) Furane, Pyridine

NC0109

2. The commercial name of trichloroethene is:

(A) Westron (B) Perclene (C) Westrosol (D) Orlone

NC0110

3. A substance containing an equal number of primary, secondary and tertiary carbon atoms is:

(A) Mesityl Oxide (B) Mesitylene (C) Maleic acid (D) Malonic acid

NC0111

4. The IUPAC name of the compound Glycerine $\underset{|}{\overset{OH}{CH_2}}-\underset{|}{\overset{OH}{CH}}-\underset{|}{\overset{OH}{CH_2}}$ is:

(A) 1,2,3-Tri hydroxy propane (B) 3-Hydroxy pentane-1,5-diol

(C) 1,2,3-Hydroxy propane (D) Propane-1,2,3-triol

NC0112

5. Which of the following is crotonic acid:

(A) CH_2= CH–COOH (B) C_6H_5–CH=CH–COOH

(C) CH_3–CH=CH–COOH (D) $\begin{matrix} CH-COOH \\ \| \\ CH-COOH \end{matrix}$

NC0113

6. The group of heterocylic compounds is:

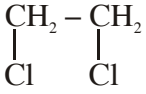
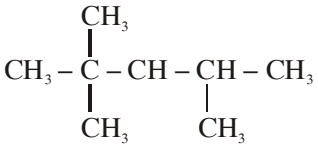
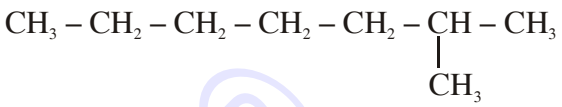
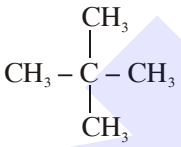
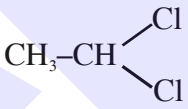
(A) Phenol, Furane (B) Furane, Thiophene

(C) Thiophene, Phenol (D) Furane, Aniline

NC0114

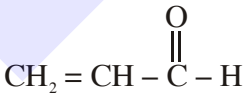
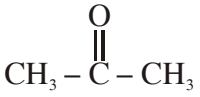
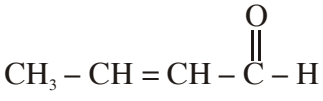
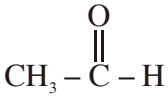
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7

Column - I (Common Name)	**Column - II (Structural formula)**
(A) Isooctane	(P) 
(B) Neopentane	(Q) 
(C) Ethylidene chloride (Geminal dihalide)	(R) 
(D) Ethylene Dichloride (Vicinal dihalide)	(S) 
	(T) 

NC0115

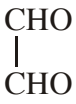
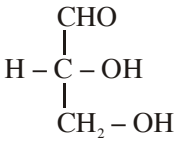
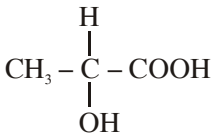
8

Column - I (Common Name)	**Column - II (Structural formula)**
(A) Acetone	(P) 
(B) Acetaldehyde	(Q) 
(C) Crotonaldehyde	(R) 
(D) Acrolein	(S) 

NC0116

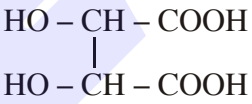
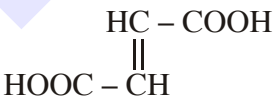
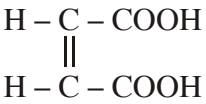
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9

Column - I (Common Name)	**Column - II (Structural formula)**
(A) 	(P) Lactic acid (In milk)
(B) 	(Q) Glyoxal
(C) $H_2N - CH_2 - COOH$	(R) Glyceraldehyde
(D) 	(S) Glycine
	(T) Glycerol

NC0117

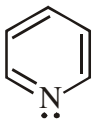
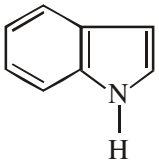
10

Column - I (Common Name)	**Column - II (Structural formula)**
(A) Fumaric acid	(P) 
(B) Adipic acid	(Q) 
(C) Maleic acid	(R) 
(D) Tartaric acid	(S) $COOH(CH_2)_4COOH$

NC0118

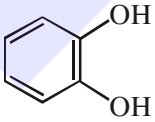
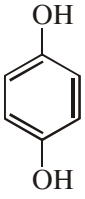
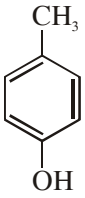
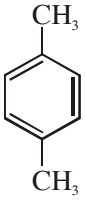
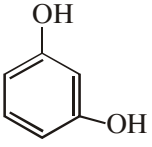
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\3-Nomen-EDE-A & B Strength-Isom-Purification\Eng\01_Nomen\01_Theory & Ex.p65

11

Column - I (Common Name)	**Column - II (Structural formula)**
(A) 	(P) Pyrrole
(B) 	(Q) Furan
(C) 	(R) Thiophene
(D) 	(S) Indol
	(T) Pyridine

NC0119

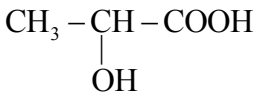
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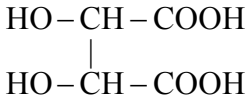
Column - I (Common Name)	**Column - II (Structural formula)**
(A) p-Cresol	(P) 
(B) p-Xylene	(Q) 
(C) Resorcinol	(R) 
(D) Quinol	(S) 
(E) Catechol	(T) 

NC0120

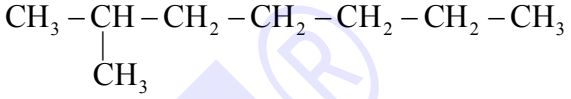
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13 Which of the following is not correctly matched:

(A) Lactic acid 

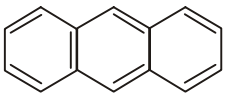
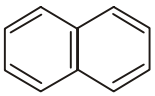
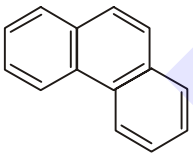
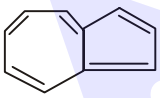
(B) Tartaric acid 

(C) Pivaldehyde $CH_3C(CH_3)_2CHO$

(D) Iso-octane 

NC0121

14

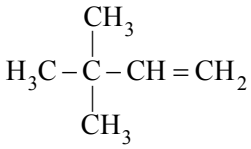
Column - I	**Column - II**
(A) 	(P) Phenanthrene
(B) 	(Q) Anthracene
(C) 	(R) Azulene
(D) 	(S) Napthalene

NC0122

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EXERCISE # JEE-ADVANCED & MAINS

1. The IUPAC name of the compound having the formula is : **[JEE 1984]**



(A) 3,3,3-trimethyl-1-propene

(B) 1,1,1-trimethyl-2-propene

(C) 3,3-dimethyl-1-butene

(D) 2,2-dimethyl-3-butene

NC0123

2. Write the IUPAC name of $CH_3CH_2CH = CH \cdot COOH$ **[JEE 1986]**

NC0124

3. The IUPAC name of the compound CH_2=CH–$CH(CH_3)_2$ is :

(A) 1,1-dimethyl-2-propene (B) 3-methyl-1-butene

(C) 2-vinyl propane (D) None of the above **[JEE 1987]**

NC0125

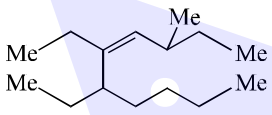
4. The number of sigma and pi-bonds in 1-butene 3-yne are: **[JEE 1989]**

(A) 5 sigma and 5 pi (B) 7 sigma and 3 pi

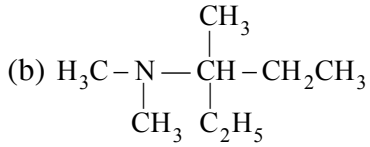
(C) 8 sigma and 2 pi (D) 6 sigma and 4 pi

NC0126

5. Write I.U.P.A.C name of following :

(a) 

Me = methyl group

(b)  **[JEE 1990]**

NC0127

6. Write IUPAC name of succinic acid. **[JEE 1994]**

NC0128

7. The IUPAC name of C_6H_5COCl is

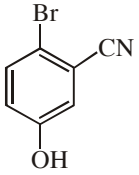
(A) Benzoyl chloride (B) Benzene chloro ketone

(C) Benzene carbonyl chloride (D) Chloro phenyl ketone **[JEE 2006]**

NC0129

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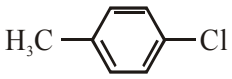
8. The IUPAC name of the following compound is **[JEE 2009]**



(A) 4-Bromo-3-cyanophenol
(B) 2-Bromo-5-hydroxybenzonitrile
(C) 2-Cyano-4-hydroxybromobenzene
(D) 6-Bromo-3-hydroxybenzonitrile

NC0130

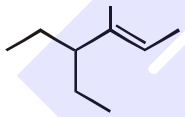
9. The IUPAC name(s) of the following compound is(are) : **[JEE 2017]**



(A) 4-methylchlorobenzene
(B) 4-chlorotoluene
(C) 1-chloro-4-methylbenzene
(D) 1-methyl-4-chlorobenzene

NC0131

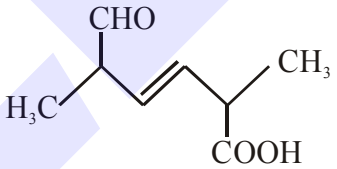
10. The IUPAC name of the following compound is : **[JEE Mains On_line 2018]**



(A) 4-methyl-3-ethylhex-4-ene
(B) 4,4-diethyl-3-methylbut-2-ene
(C) 3-ethyl-4-methylhex-4-ene
(D) 4-ethyl-3-methylhex-2-ene

NC0132

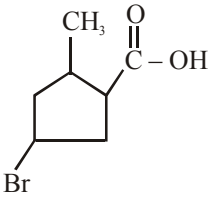
11. The IUPAC name for the following compound is: **[JEE MAINS 2020]**



(1) 2, 5-dimethyl-6-carboxy-hex-3-enal
(2) 6-formyl-2-methyl-hex-3-enoic acid
(3) 2, 5-dimethyl-5-carboxy-hex-3-enal
(4) 2, 5-dimethyl-6-oxo-hex-3-enoic acid

NC0133

12. The IUPAC name of the following compound is : **[JEE MAINS 2020]**

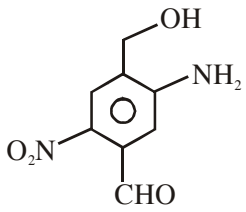


(1) 4-Bromo-2-methylcyclopentane carboxylic acid
(2) 5-Bromo-3-methylcyclopentanoic acid
(3) 3-Bromo-5-methylcyclopentane carboxylic acid
(4) 3-Bromo-5-methylcyclopentanoic acid

NC0134

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13. The IUPAC name of the following compound is : **[JEE MAINS 2020]**



(1) 3-amino-4-hydroxymethyl-5-nitrobenzaldehyde

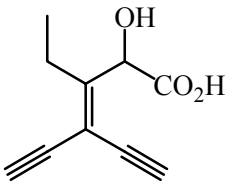
(2) 2-nitro-4-hydroxymethyl-5-aminobenzaldehyde

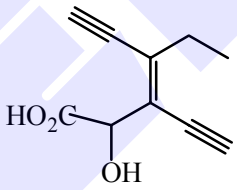
(3) 4-amino-2-formyl-5-hydroxymethylnitrobenzene

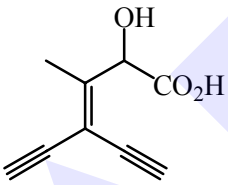
(4) 5-amino-4-hydroxymethyl-2-nitrobenzaldehyde

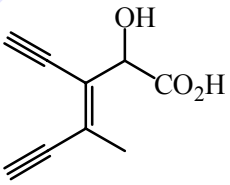
NC0135

14. Which one of the following structures has the IUPAC name **[IIT-2020]**
3-ethynyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid ?

(A) 

(B) 

(C) 

(D) 

NC0136

15. In $\overset{1}{CH_2}=\overset{2}{C}=\overset{3}{C}H-\overset{4}{C}H_3$ molecule, the hybridization of carbon 1,2,3 and 4 respectively are :

(1) sp^3, sp, sp^3, sp^3 (2) sp^2, sp^2, sp^2, sp^3 **[JEE MAINS 2021]**

(3) sp^2, sp, sp^2, sp^3 (4) sp^2, sp^3, sp^2, sp^3

NC0137

16. The number of sigma bonds in **[JEE MAINS 2021]**

$H_3C-\underset{\underset{H}{|}}{C}=CH-C\equiv C-H$ is _____ . **NC0138**

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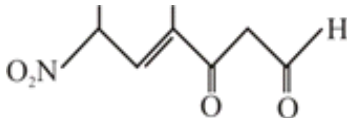
17. The IUPAC name of ethylidene chloride is :- **[JEE MAINS 2022]**

(1) 1-Chloroethene (2) 1-Chloroethyne

(3) 1,2-Dichloroethane (4) 1,1-Dichloroethane

NC0139

18. The correct IUPAC name of the following compound is : **[JEE MAINS 2022]**

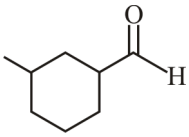


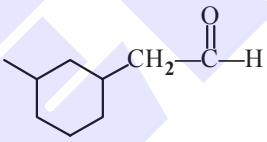
(1) 4-methyl-2-nitro-5-oxohept-3-enal (2) 4-methyl-5-oxo-2-nitrohept-3-enal

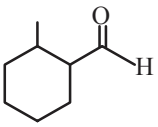
(3) 4-methyl-6-nitro-3-oxohept-4-enal (4) 6-formyl-4-methyl-2-nitrohex-3-enal

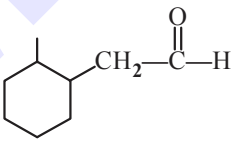
NC0140

19. Correct structure of γ-methylcyclohexane carbaldehyde is : **[JEE MAINS 2022]**

(1) 

(2) 

(3) 

(4) 

NC0141

20. The correct decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is : **[JEE MAINS 2022]**

(1) –COOH > $–CONH_2$ > –COCl > –CHO

(2) $–SO_3H$ > –COCl > $–CONH_2$ > –CN

(3) –COOR > –COCl > $–NH_2$ > $>C=O$

(4) COOH > COOR > $CONH_2$ > COCl

NC0142

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ANSWER-KEY

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	D	D	C	B	A	D	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	D	C	B	C	D	B	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	B	D	C	D	D	C	B	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	B	B	D	C	B	C	C	B	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	D	B	D	B	B	A	B	D	D
Que.	51	52	53							
Ans.	C	A	D							

EXERCISE # O-II

1 4-Ethyl octane

2 3-Ethyl-2,4-dimethyl pentane

3 5-Methyl hepta-1,3,6-triene

4 Hepta-1,5-dien-3-yne

5 2-Isopropyl-4-methyl pent-1-ene

or 4-Methyl-2-(methyl ethyl) pent-1-ene

6 3-Methoxypropene

7 1-Hydroxybut-3-en-2-one

8 2-Ethylbut-2-en-1-ol

9 3-nitroprop-2-en-1-ol

10 4-hydroxyhex-5-en-1-yn-3-one

11 4,6-Bis-[1,1-Dimethyl ethyl] Nonane

12 2-Formyl pentane nitrile

13 2,2,6,7-tetramethylocatane

14 3-Ethyl-4,6-dimethyloctane

15 5-Methyl cyclohexa-1,3-diene

16 4-Ethyl Pent-4-en-2-amine

17 1,2-epoxy propane

18 1,3,4-trimethyl cyclobutene

19 Methylene cyclohexane

20 1-ethyl-2-methylcyclopentane

21 1-methyl-3-(methyl ethyl) cyclohexane or 1-isopropyl-3-methylcyclohexane

22 Butyl cyclohexane

23 Isopropylidenecyclopentane or 1-methyl ethylidene cyclopentane

24 3-Bromo-4-cyclopropyl cyclopentane carboxylic acid

25 Cyclopent-2-en-1-one

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26 1-(3-butenyl) cyclopentene

27 1,2-diethenyl cyclohexene

28 1-cyclohexyl-1-propanone

29 Ethyl cyclohexanecarboxylate

30 4-Bromo-2-ethyl cyclopentanone

31 3-(hydroxymethyl)-5-methylheptanal

32 2-Bromo-6-oxocyclohexanecarbaldehyde

33 5-amino-6-(1-methyl propyl) cyclo hex-2-enol

34 2-bromo-2-methyl cyclopentanone

35 Methyl-2-methoxy-6-methyl-3- cyclohexene carboxylate

36 Bicylo(2,2,1)heptane

37 9-methyl bicyclo(4,2,1) nonane

38 Bicyclo [3,2,2] Non-6-one

39 spiro(4,5) decane

40 2-Methyl Benzoyl Chloride

41 1,3,3-Trimethyl cyclohexene

42 Bicyclo(2,2,1) heptane

43 8-chloro bicyclo(4,2,0) oct-2-ene

44 2-cyclopenten-1-ol

45 Ethyl-2-oxo cyclo pentane carboxylate

46 2-Formyl Benzoic acid

47 3-Mthyl Benzoic acid

48 Cyclohex-2-en-1,4-dione

49 2-ethynyl cyclohexanol

50 4-chloro-1-cyclopentyl pentane-2-one

51 1-Amino methyl-2-ethyl cyclohexanol

52 4-isopropyl -1-propyl cyclohexene or 4-(methyl ethyl)-1-propyl cyclohexene

53 2-(2-oxo-cyclohexyl) propanoic acid

54 3-ethoxy-1(1-nitrocyclohexyl)-hex-4-en-1-one

55 1,3-diphenyl-1,4-pentadiene

EXERCISE # S-I

Que.	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Ans.	D	C	B	D	C	B	A→Q; B→S; C→T; D→P
Que.	**8**			**9**			**10**
Ans.	A→Q; B→S; C→T; D→P			A→Q; B→R; C→S; D→P			A→Q; B→S; C→R; D→P
Que.	**11**				**12**		**13**
Ans.	A→T; B→P; C→S; D→Q				A→R; B→S; C→T; D→Q; E→P		D
Que.	**14**						
Ans.	A→Q; B→S; C→P; D→R						

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EXERCISE # JEE-ADVANCED & MAINS

1. (C)

2. $\overset{5}{CH_3}-\overset{4}{CH_2}-\overset{3}{CH}=\overset{2}{CH}-\overset{1}{COOH}$

2-pentene-1-oic acid and or 2-pentenoic acid

3. (B) **4. (B)**

5. (a) 5,6-diethyl-3-methyl-dec-4-ene

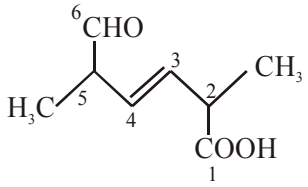
(b) N,N, 3-trimethyl-3-pentanamine

6. Butane-1,4-dioic acid

7. (C) **8. (B)**

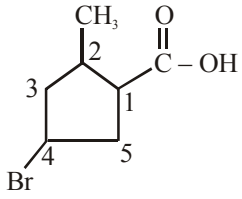
9. (B,C) **10. (D)**

11. (4)

Sol.  IUPAC name

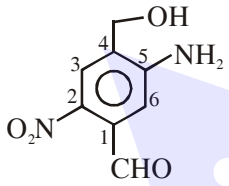
2, 5-dimethyl-6-oxo-hex-3-enoic acid

12. (1)

Sol. 

4-bromo-2-methyl cyclopentane carboxylic Acid

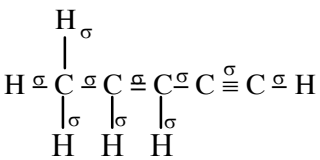
13. (4)

Sol. 

5-amino-4-hydroxymethyl-2-nitrobenzaldehyde

14.(D) **15. (3)**

16. (10)

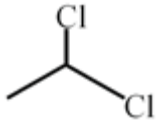
Sol. 

numbers of bonds = 10

ALLEN®

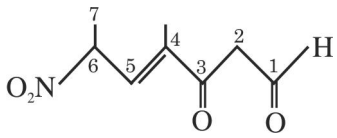
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17. (4)

Sol. 

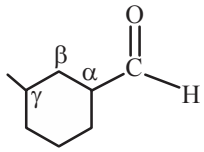
"1, 1-Dichloroethane is Ethylidene chloride"

18. (3)

Sol. 

4-Methyl-6-nitro-3-oxohept-4-enal

19. (1)

Sol. 

γ-methyl cyclohexane carbaldehyde

20. (2)

Sol. $-SO_3H > -COCl > -CONH_2 > -CN$

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Important Notes

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